

Technical Datasheet

VBF-T4R2-DSH-01 | case t4_r2

Extreme-cold equipment cell · finalist E3-porheadroom · Chen2020 base (NMC811-class/graphite) · 2026-08-26

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Field	Value	Source
Rated capacity	5.0 Ah nominal; 5.0392 Ah simulation-verified (DFN 1C, 25 °C, +0.78 %)	parameter set + cell/r5_E3_1c_dfn.json
Nominal voltage / window	midpoint 3.847 V (1C discharge); window 2.5 – 4.2 V	r5_E3_energy_dfn.json midpoint_voltage_v; parameter set
Rated energy	18.16 Wh	r5_E3_energy_dfn.json energy_wh (V-I time integration)
Energy density	459.45 Wh/kg (contract caliber, electrolyte excluded from mass)	r5_E3_energy_dfn.json
Volumetric energy density	936.82 Wh/L (Σ layer thickness × area)	r5_E3_1c/energy_dfn.json
Maximum continuous discharge rate	1C verified at 25 °C (5.0392 Ah) and at –20 °C (5.0078 Ah)	r5_E3_1c_dfn.json, r5_E3_lowt_dfn.json
Fast-charge capability	4C @ 45 °C: no plating (anode potential min +0.0060 V), T_max 326.96 K (≤ 333.15, margin 6.19 K); 0.589 Ah accepted in 4C leg	r5_E3_4c_dfn2.json
Operating temperature range	–20 °C discharge verified (cold-soak protocol); 45 °C charge verified. Outside this range: not simulated — honest simulation boundary	protocol conditions recorded
Cycle life	Not simulated (requires aging model) — not fabricated	template-required honest statement
Safety determination	PASS: T_max 326.96 K ≤ 333.15 K; plated = false (0/307 anode-potential points negative)	r5_E3_4c_dfn2.json (stage-4 exam)
DCR (informational)	2.07 mΩ at 10 % discharge (start-OCV minus V _{10%} ÷ I _{1C})	r5_E3_energy_dfn.json dcr_ohm
Dimensions and mass	electrode strip 65.0 mm × 1580.0 mm × 0.1888 mm stack; mass 39.54 g (layers, electrolyte excluded) + 6.01 g electrolyte = 45.55 g; wound outer dimensions and enclosure Not provided	parameter set + calc-energy + pore-volume formula
Electrolyte (formulation estimates)	σ 1.6 S/m, D 3.2e-10 m ² /s, t ⁺ 0.6 — blended single-ion class estimates, temperature-independent (like the set's flat functions), not measured	parameter overrides, R4/R5 evaluate reasoning
Cell chemistry notes	NMC811-class positive (r 3.5 μm, por 0.28), graphite-class negative (r 3.0 μm, por 0.28), separator 8 μm, collectors Al 12 μm / Cu 8 μm, N/P = 1.03 (positive-limited)	parameter set + N/P derivation script